

# Benchmarking of NQCH's quantum computer

September 27, 2026

## 1. Report of Changes

**Platform:** sinq20

**Calibration-id:** ae51b7967ef92463a98d705f04fb00e9d9d10190

**Calibration date:** 2026-08-12 02:52:08

**Calibration note:** chore(sinq20): q5 sq recal

**Experiment-id:** 20260927212203

**Experiment date:** 2026-09-27 21:22:04

**Experiment note:** temporary note!!!

**Platform:** sinq20

**Calibration-id:** 4dc4082f38a53222b3956c22202d32a520d4bc78

**Calibration date:** 2026-01-15 02:03:03

**Calibration note:** chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

**Experiment-id:** 20260118161538

**Experiment date:** 2026-01-18 16:15:38

**Experiment note:** temporary note!!!

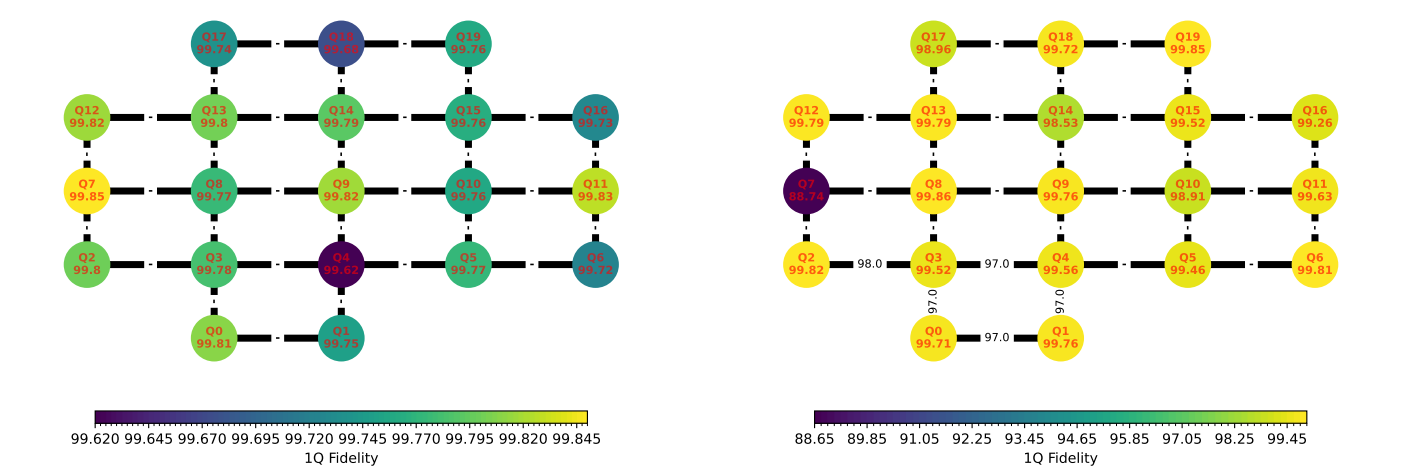
## 2. Version Comparison

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.3.3   | numpy   | 2.5.3   |
| qibolab      | 0.2.17  | qibocal | 0.2.7   |
| matplotlib   | 3.11.1  | scipy   | 1.18.1  |
| scikit-learn | 1.9.0   | pandas  | 3.0.5   |
| networkx     | 3.6.1   | sympy   | 1.14.0  |
| torch        | 2.14.0  |         |         |

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.22  | numpy   | 2.2.6   |
| qibolab      | 0.2.9   | qibocal | 0.2.4   |
| matplotlib   | 3.10.3  | scipy   | 1.15.3  |
| scikit-learn | 1.6.1   | pandas  | 2.2.3   |
| networkx     | 3.4.2   | sympy   | 1.14.0  |
| torch        | 2.7.0   |         |         |

## 3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



#### 4. Statistics

|                | Average | Median | Min | Max |
|----------------|---------|--------|-----|-----|
| T1 (ns)        | -       | -      | -   | -   |
| T2 (ns)        | -       | -      | -   | -   |
| Fidelity       | -       | -      | -   | -   |
| RO Fidelity    | -       | -      | -   | -   |
| 2q RB Fidelity | -       | -      | -   | -   |
| 2q CZ Fidelity | -       | -      | -   | -   |

|                | Average  | Median   | Min   | Max      |
|----------------|----------|----------|-------|----------|
| T1 (ns)        | 1.28e+04 | 1.23e+04 | 646   | 3.65e+04 |
| T2 (ns)        | 2.36e+25 | 3.68e+03 | 125   | 9.43e+26 |
| Fidelity       | 0.99     | 0.997    | 0.887 | 0.999    |
| RO Fidelity    | 0.794    | 0.777    | 0.777 | 0.927    |
| 2q RB Fidelity | 0.965    | 0.972    | 0.923 | 0.981    |
| 2q CZ Fidelity | 0.975    | 0.974    | 0.967 | 0.991    |

#### 5. Best Qubits Selection

| k-qubits | Best Qubits | Fidelity |
|----------|-------------|----------|
| 2        | N/A         | N/A      |
| 3        | N/A         | N/A      |
| 4        | N/A         | N/A      |
| 5        | N/A         | N/A      |

| k-qubits | Best Qubits   | Fidelity |
|----------|---------------|----------|
| 2        | 2, 3          | 0.981    |
| 3        | 0, 2, 3       | 0.976    |
| 4        | 0, 1, 2, 3    | 0.970    |
| 5        | 0, 1, 2, 3, 4 | 0.965    |

#### 6. Summary metrics

| Benchmark | Metric | Value | Runtime |
|-----------|--------|-------|---------|
| Mermin    | Max    | -     | -       |

| Benchmark | Metric | Value  | Runtime  |
|-----------|--------|--------|----------|
| Mermin    | Max    | -3.316 | 1.02 sec |

#### 7. Randomized Benchmarking Results

| Qubit n | Fidelity | Error Bars     | Qubit n | Fidelity | Error Bars     |
|---------|----------|----------------|---------|----------|----------------|
| 0       | 0.998    | $\pm 8.62e-05$ | 10      | 0.998    | $\pm 0.000103$ |
| 1       | 0.997    | $\pm 0.000103$ | 11      | 0.998    | $\pm 6.63e-05$ |
| 2       | 0.998    | $\pm 9.66e-05$ | 12      | 0.998    | $\pm 6.58e-05$ |
| 3       | 0.998    | $\pm 0.000153$ | 13      | 0.998    | $\pm 0.000102$ |
| 4       | 0.996    | $\pm 0.000226$ | 14      | 0.998    | $\pm 8.09e-05$ |
| 5       | 0.998    | $\pm 0.000124$ | 15      | 0.998    | $\pm 0.000103$ |
| 6       | 0.997    | $\pm 0.000121$ | 16      | 0.997    | $\pm 0.000162$ |
| 7       | 0.998    | $\pm 4.57e-05$ | 17      | 0.997    | $\pm 8.82e-05$ |
| 8       | 0.998    | $\pm 0.000107$ | 18      | 0.997    | $\pm 0.000138$ |
| 9       | 0.998    | $\pm 0.000117$ | 19      | 0.998    | $\pm 8.11e-05$ |

| Qubit n | Fidelity | Error Bars     | Qubit n | Fidelity | Error Bars     |
|---------|----------|----------------|---------|----------|----------------|
| 0       | 0.997    | $\pm 0.000131$ | 10      | 0.989    | $\pm 0.000582$ |
| 1       | 0.998    | $\pm 7.47e-05$ | 11      | 0.996    | $\pm 0.000218$ |
| 2       | 0.998    | $\pm 7.57e-05$ | 12      | 0.998    | $\pm 0.000126$ |
| 3       | 0.995    | $\pm 0.000163$ | 13      | 0.998    | $\pm 7.15e-05$ |
| 4       | 0.996    | $\pm 0.000185$ | 14      | 0.985    | $\pm 0.00183$  |
| 5       | 0.995    | $\pm 0.000147$ | 15      | 0.995    | $\pm 0.000469$ |
| 6       | 0.998    | $\pm 8.34e-05$ | 16      | 0.993    | $\pm 0.000558$ |
| 7       | 0.887    | $\pm 0.0287$   | 17      | 0.99     | $\pm 0.000502$ |
| 8       | 0.999    | $\pm 4.48e-05$ | 18      | 0.997    | $\pm 0.00026$  |
| 9       | 0.998    | $\pm 7.2e-05$  | 19      | 0.998    | $\pm 9.05e-05$ |

8. Mermin

| Runtime | Qubits | Max |
|---------|--------|-----|
| -       | -      | -   |

| Runtime  | Qubits  | Max    |
|----------|---------|--------|
| 1.02 sec | 0, 2, 3 | -3.316 |

